

DAC Module (VMIVME-4140)

DAC Module (Digital to Analog Converter VMIVME4140)

The VMIVME-4140 Analog Output Board has 16 analog output channels with 12-bit resolution, sourcing up to 10mA at $\pm 10V$. Each output has a dedicated Digital to Analog Converter (DAC) assigned to it. The outputs may be configured as unipolar(0 to $\pm 10V$, 0 to $\pm 5V$, 0 to $\pm 2.5V$) or bipolar($\pm 2.5V$, $\pm 5V$, $\pm 10V$).

Configuring of DAC for operation.

There are two options to configuring logic of operation of PLU. One of them is used routines from 'vmic4140.o' library(located on adaql2: "/home/moller/dev/vmic4140/") in the coda '*.crl' file. The second one is remote configuration using 'plu_server' running on VME CPU and client program 'cli.pl' (perl script) (located in adaql2:~/dev/vx/plu_server/).

To configuring the logic of the module operation next functions from 'vmic4140.o' library are used:

init_4140() - performs allocation of address and space for DAC register on VME bus and set default mode of operation: $\pm 10V$, online, 2's compliment, no sync

Parameters: none;

Returns: 0 - on successfull and -1 otherwise.

config_4140(data) - configures mode of operation of DAC

Parameters: data - configuration word(16bit) to write to Configuration Control Register (Offset \$002)

Returns: none

set_out_4140(channel, value) - load 'value' to DAC register for selected 'channel'

Parameters: channel - integer (0-15) channel number, value - integer (0x0000-0x0FFF) value for DAC register *Returns:* 0 on successfull and -1 otherwise.

get_out(channel) - returns value from DAC register for selected 'channel'.

Parameters: channel - channel number(0-15);

Returns: value from DAC register

Examples.

Before using the any functions 'init_4140()' should be executed.

1. To configuring DAC for operation: range: $\pm 10V$, 2's compliment, no sync:

```
config_4140(0xC000);
```

2. Load value '1000' to DAC register for channel=0:

```
set_out_4140(0, 1000);
```

Remote configuration.

For remote control of DAC outputs the perl script 'cli.pl' or 'hvcli' programm are used.

Format of configuration command for script 'cli.pl' is next:

```
cli.pl host port "message"
```

where:

host -is hostname (halladaq10.jlab.org) of the VME CPU where 'plu_server' is running

port -is port number (1357) to connect to the 'plu_server'.

"message" -is command string(in double or single quotas) to send to the 'plu_server'.

The message format of command string is next:

"destination command [parameter] [value1] [value2]"

where:

destination - module name from next list : PLU, TDC, DAC, VME, SERVER

command - command name: SET, CONFIG, GET, INIT, RESET, WRITE, READ

parameter - parameter name: AND, OR, NOT, TEST, OUT, STATUS

The format of the respond message is next if command is successfully executed :

"OK <sent command message> [return value]"

On unsuccessfully execution the respond message is:

"ERROR <error description>"

The 'plu_server' supports next commands for DAC: INIT, SET, GET, TEST
and list parameters: OUT

DAC commands description:

INIT - initialization of DAC;

CONFIG - setting of operation mode of DAC;

GET - get the data from output register of DAC.

SET - load data to output register of DAC;

Examples:

Command message for setting of operation mode of DAC (range:+/-10V, 2's compliment, no sync) :

`cli.pl halladaq10 1357 "DAC config 0xC000"`

response message:

"OK DAC CONFIG AND 0xC000"

Command message for load value=256 to the channel=1:

`cli.pl halladaq10 1357 "DAC SET 1 256"`

response message:

"OK DAC SET 1 256"

or

`cli.pl halladaq10 1357 "dac set 1 0x00FF"`

response message:

"OK DAC SET 1 0x00FF"

Get data from output register of DAC for output(channel) #4 :

`cli.pl halladaq10 1357 "dac GET 4"`

response message:

"OK DAC GET 4 1024"
